

CLIENT REPORT

Crash Modeling

Support safer roads by identifying high-risk locations and factors associated with elevated crash risk.

REPORT REFERENCE

AUD-8F3965C55C

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BÓVEDA VERSION

Alpha 1.0.2

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The project and Bóveda's position

A concise view of the project, its primary recorded result and the parts of the record that need attention.

What the project does

The project addresses urban road safety by estimating and mapping crash risk for individual road segments using crash history, road attributes and public concerns. City transportation departments are intended to use these outputs to identify higher-risk locations and inform interventions, although the displayed evaluation covers a filtered Boston benchmark only.

Evidence E-B335630551 · E-3102C1ACD6 · E-C998E295B4

Best recorded result

ROC AUC

0.735536

Model

XGBoost classifier using five features

Target

Any crash on a segment in week 19 of 2017

Evaluation

1,026 held-out records from a random split

Evidence E-3102C1ACD6

Partial reconstruction

Bóveda could trace the main result to the data, model and evaluation sample behind it. However, 2 important applicable analytical areas could not be reconstructed, so the overall reconstruction remains partial.

What needs attention

1

Signal

1 High

5

Evidence Gaps

Overall evidence position

The available project record supports some areas more completely than others, but exact reproduction remains incomplete because 5 evidence gaps are still unresolved.

Coverage not established

PROJECT EXPLANATION

Project overview

What the project is intended to do, the objects it analyses and the use it is intended to support.

Purpose

The project addresses urban road safety by estimating and mapping crash risk for individual road segments using crash history, road attributes and public concerns. City transportation departments are intended to use these outputs to identify higher-risk locations and inform interventions, although the displayed evaluation covers a filtered Boston benchmark only.

Evidence E-B335630551 · E-3102C1ACD6 · E-C998E295B4

Analytical task

Estimate and map crash risk for each road segment using historical and segment-level features.

INTERPRETED_INFERRED

Evidence E-B335630551 · E-E1D5879CE3

Target / outcome

A probability or binary prediction of a crash on each road segment.

OBSERVED

Evidence E-B335630551 · E-3102C1ACD6

Unit of observation

One Boston road segment for the target week

INTERPRETED_INFERRED

Evidence E-3102C1ACD6

Population / scope

Road segments in cities supplying geocoded crash data; the evaluated benchmark is Boston-specific.

INTERPRETED_INFERRED

Evidence E-B335630551 · E-3102C1ACD6

Intended use

City transportation departments use segment risk maps to prioritize locations and inform timely safety interventions.

OBSERVED

Evidence E-B335630551 · E-C998E295B4

Project period

Data through week 19 of 2017; concerns summarized for 2016

OBSERVED

Evidence E-3102C1ACD6

Major data sources

Boston crash records, road data, Vision Zero concerns, traffic recordings and turning counts.

OBSERVED

Evidence E-B335630551 · E-3102C1ACD6

DATA & POPULATIONS

What data the project operated on

Only supervisory questions answered by the available evidence are included below.

Data in brief

Boston segment records were filtered through road, traffic and concern joins; the benchmark then formed five crash-history and concern features for a one-week binary target.

Evidence E-3102C1ACD6

Source data	22,466 Boston road segments DERIVED Evidence E-B7DD9F89DA
Model sample	2,283 training records OBSERVED Evidence E-3102C1ACD6
Evaluation sample	1,026 test records OBSERVED Evidence E-3102C1ACD6
Main population filters	Segments required a historical crash and successful joins to segment, traffic-recording and concern data before the benchmark split. OBSERVED Evidence E-3102C1ACD6
Known population limitation	Excluding segments without historical crashes removes much of the road network, limiting conclusions about lower-history or previously crash-free segments. INTERPRETED_INFERRED Evidence E-B7DD9F89DA · E-3102C1ACD6



Model-ready population

3,309



Training data

2,283



Evaluation data

1,026



Population / sample lineage

How the original population became the training and evaluation samples.

Original data

→ filter →

Model-ready population

Segments required a historical crash and successful joins to segment, traffic-recording and concern data before the benchmark split.

Evidence E-B7DD9F89DA · E-3102C1ACD6

Model-ready population

→ split →

Training data

Evidence E-3102C1ACD6

Model-ready population

→ split →

Evaluation data

Evidence E-3102C1ACD6

What was recorded, and what it supports

The strongest comparable final result is shown with its evaluation context and material limitation.

Best recorded result

ROC AUC

0.735536

Model

XGBoost classifier using five features

Target

Any crash on a segment in week 19 of 2017

Evaluation sample

1,026 held-out records from a random split

Evidence E-3102C1ACD6

Evaluation design

A random 70% training split supported five-fold randomized tuning; final ROC AUC was measured on the remaining 1,026 records.

OBSERVED

Evidence E-3102C1ACD6

Principal limitation

The benchmark excludes segments without recorded crashes and requires several successful data joins, restricting interpretation beyond this selected subset.

INTERPRETED_INFERRED

Evidence E-3102C1ACD6

Other recorded material results

These outcomes retain their own target and evaluation context. They are not treated as directly comparable unless a model comparison below establishes a shared setup.

RESULT	METHOD / TARGET	EVALUATION CONTEXT	EVIDENCE
Brier score 0.148798	XGBoost classifier using five features Crash probability for week 19 of 2017	Same 1,026-record held-out benchmark test set	OBSERVED Evidence E-3102C1ACD6
F1 score 0.059072	XGBoost classifier using five features Binary crash classification for week 19 of 2017	Same 1,026-record held-out benchmark test set	OBSERVED Evidence E-3102C1ACD6

What the result establishes

On the held-out filtered benchmark, XGBoost distinguished segment records with and without a target-week crash at 0.73553583169 ROC AUC.

Evidence E-3102C1ACD6

What it does not establish

It does not show performance across all city segments or that acting on risk scores prevents crashes.

Evidence E-B7DD9F89DA · E-3102C1ACD6

Model comparison

Any crash on a segment in week 19 of 2017

Results recorded for the same target and evaluation setup.

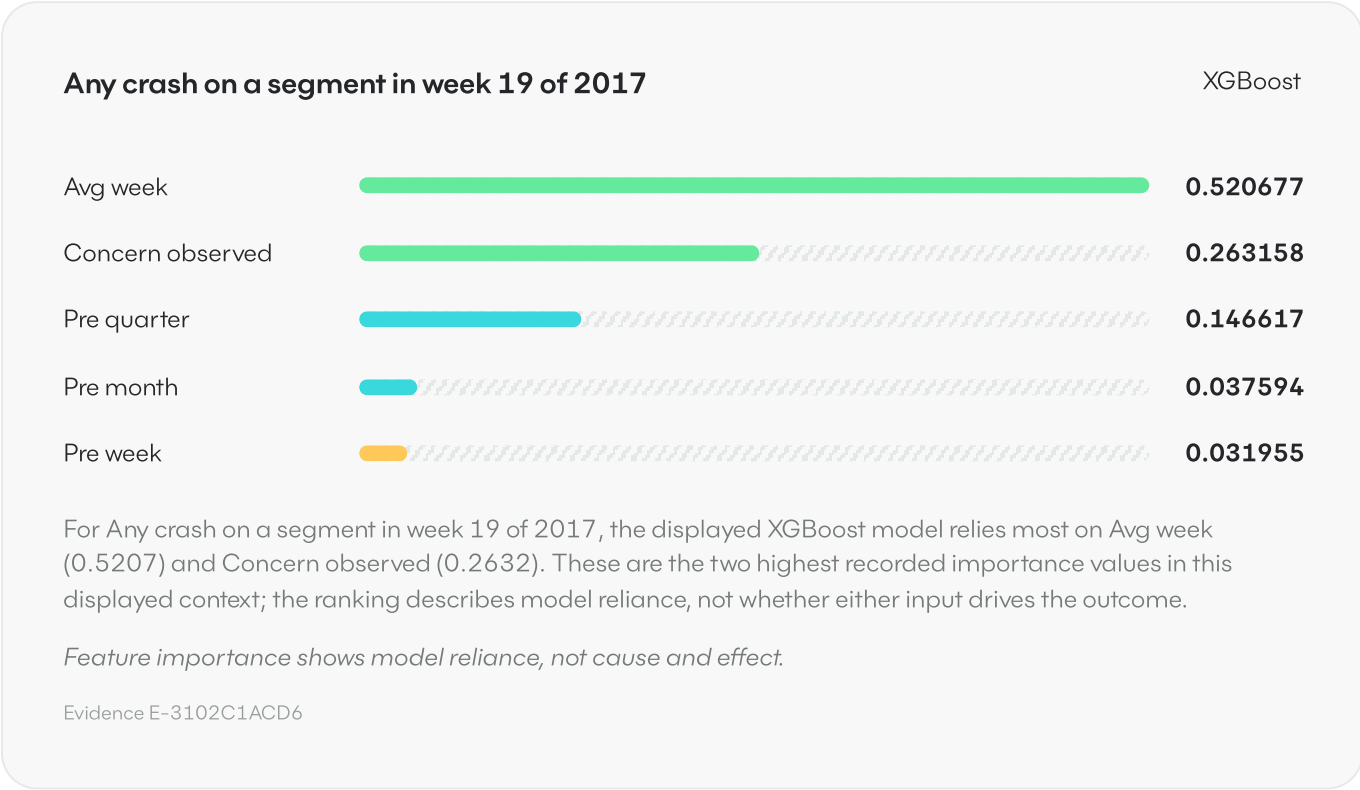
METRIC	RANDOM FOREST	LOGISTIC REGRESSION	XGBOOST
ROC AUC	0.51	0.507	0.736
Brier loss	0.083	0.215	0.149
F1	0.081	0.043	0.059

XGBoost records the strongest ROC AUC (0.7355), while Random Forest leads on Brier loss (0.0826). No single model leads across every available metric, and the rows should be interpreted separately because they measure different aspects of performance.

FEATURE / DRIVER EVIDENCE

What the displayed model relied on most

The ranking describes relative model reliance within one context; it does not establish cause and effect.



What Bóveda found

Signals describe material project conditions detected by deterministic checks. They do not automatically mean errors, failures, violations or compliance problems.

01

High

2 affected analytical areas

Important analytical evidence is missing

What Bóveda found

Important analytical evidence is entirely absent in 2 applicable areas, preventing meaningful supervisory assessment of each one.

Why it matters

The missing evidence prevents Bóveda from meaningfully assessing an important part of the project, rather than merely limiting a detail within an otherwise assessable area.

Supported by 2 verification checks

Evidence E-3102C1ACD6

Where the record is incomplete

These are parts of the project that Bóveda could not fully verify or reproduce from the available evidence. A gap does not mean something is wrong; it means the project record is incomplete.

01

Evidence Gap

Reproducibility

The exact data version used for the result is not fully identified

What Bóveda found

Bóveda found the project data, but not enough information to identify the precise dataset snapshot behind the reported result.

Why it matters

A precise data version is needed to reproduce the same analysis from the same inputs.

Supported by 1 verification check

Evidence E-00147FE2AC · E-1BE4DCDB4E · E-2D321D0CCB · E-3102C1ACD6 · E-33DC416CAE · E-38DA02CA70 · E-3980CB24A2 · E-3D83962C95 · E-49694FAF5B · E-4D7C51B1EF · E-68417CB343 · E-784758FCB5 · E-B7DD9F89DA · E-C341F43CFC · E-C6EF786BDD · E-DD2C0EB6EA · E-E0D7C4E878 · E-E20AB3DB80 · E-E59BA1BCCB · E-EC0B19537E · E-F313830786

The reported metric cannot be traced through its complete calculation

What Bóveda found

Bóveda found the reported metric, but not a complete link to the calculation and score inputs that produced it.

Why it matters

Without that calculation trail, Bóveda cannot fully verify how the reported value was obtained.

Supported by 1 verification check

Evidence E-00147FE2AC · E-1BE4DCDB4E · E-2D321D0CCB · E-3102C1ACD6 · E-33DC416CAE · E-38DA02CA70 · E-3980CB24A2 · E-3D83962C95 · E-49694FAF5B · E-4D7C51B1EF · E-68417CB343 · E-784758FCB5 · E-B7DD9F89DA · E-C341F43CFC · E-C6EF786BDD · E-DD2C0EB6EA · E-E0D7C4E878 · E-E20AB3DB80 · E-E59BA1BCCB · E-EC0B19537E · E-F313830786

Bóveda cannot fully link the predictions to the labels used to evaluate them

What Bóveda found

The available project record does not contain enough information to reconstruct the complete link between predictions and evaluation labels.

Why it matters

Without that link, Bóveda cannot fully reproduce or independently verify the reported evaluation result.

Supported by 1 verification check

Evidence E-00147FE2AC · E-1BE4DCDB4E · E-2D321D0CCB · E-3102C1ACD6 · E-33DC416CAE · E-38DA02CA70 · E-3980CB24A2 · E-3D83962C95 · E-49694FAF5B · E-4D7C51B1EF · E-68417CB343 · E-784758FCB5 · E-B7DD9F89DA · E-C341F43CFC · E-C6EF786BDD · E-DD2C0EB6EA · E-E0D7C4E878 · E-E20AB3DB80 · E-E59BA1BCCB · E-EC0B19537E · E-F313830786

Evaluation behaviour cannot be assessed

What Bóveda found

Bóveda found an applicable classification result, but no linked diagnostic evidence showing false positives, false negatives or other behaviour behind the headline score.

Why it matters

Without diagnostic evidence, a supervisor cannot see important error patterns that may be hidden by the headline metric.

Supported by 1 verification check

Evidence E-3102C1ACD6

Missing-data handling cannot be assessed

What Bóveda found

Bóveda found missing-value handling in the focal analytical workflow, but no stage- and denominator-aware measurements showing how much data was missing.

Why it matters

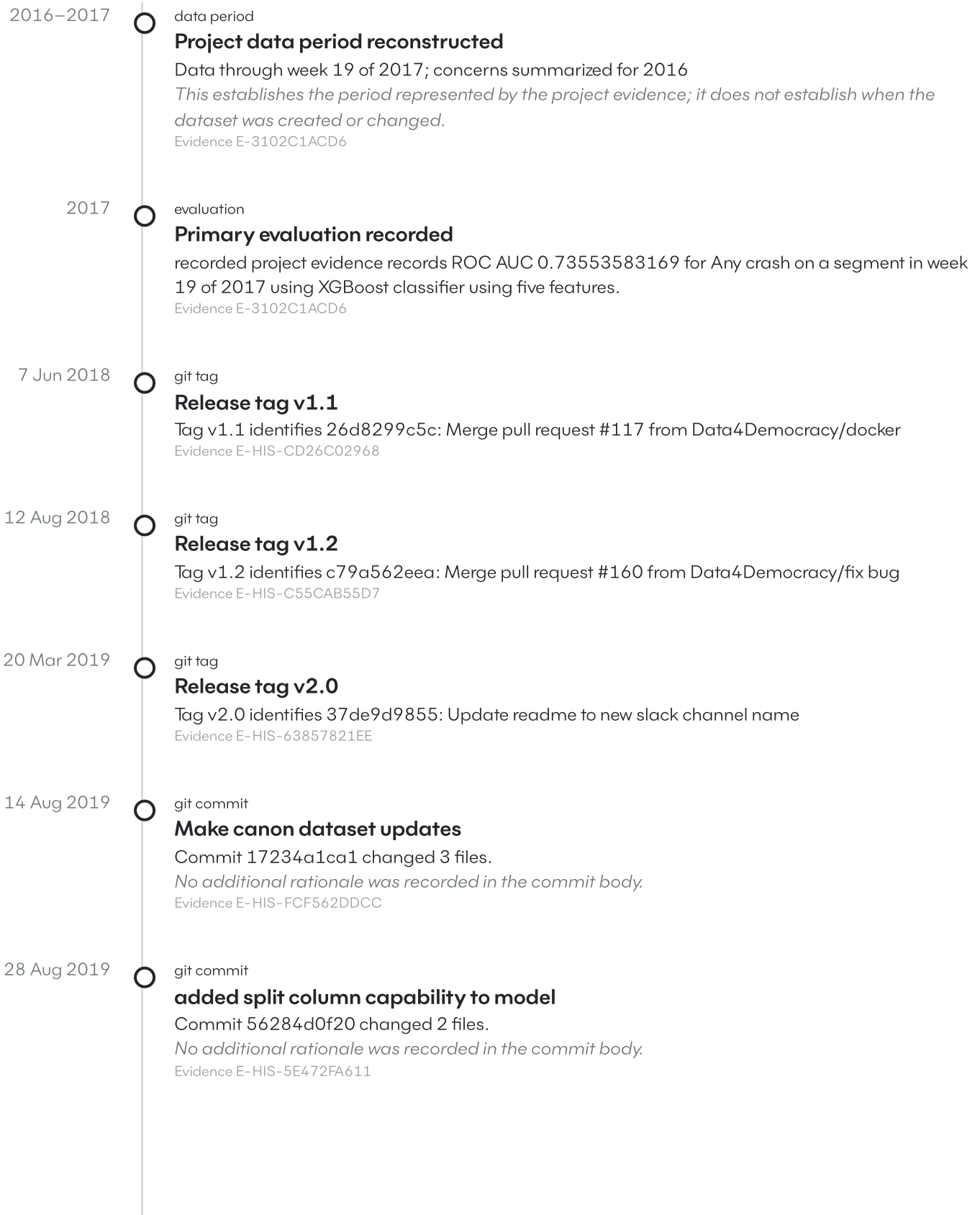
Without those measurements, a supervisor cannot assess the scale of missingness or how much the handling step affected the analytical data.

Supported by 1 verification check

Evidence E-3102C1ACD6

Material events in the available record

Bóveda found extensive recorded project history. This view elevates 12 material host-project events and retains 294 other changes as supporting context; older changes remain outside the bounded collection.



9 Sept 2019	git commit Update make preds viz + pipeline to generate postprocessed files for split columns Commit 434780a182 changed 2 files. <i>No additional rationale was recorded in the commit body.</i> Evidence E-HIS-A19507381C
8 Oct 2019	git commit Remove risk map from pipeline Commit 6b22e7085c changed 1 file. <i>No additional rationale was recorded in the commit body.</i> Evidence E-HIS-574EA43B97
20 Dec 2019	git commit Added new city config for NYC. Ran pipeline and tested the web app, works but does not show modes yet.. Commit 6edc5c8c92 changed 1 file. <i>No additional rationale was recorded in the commit body.</i> Evidence E-HIS-EFAA393053
25 Jun 2020	git commit fix to model use appropriate featureset Commit 9a5203a5ec changed 1 file. <i>No additional rationale was recorded in the commit body.</i> Evidence E-HIS-6E325D0345
30 Dec 2021	git commit adaptation for multi feature doc fixes and segment format fix Commit 9fb8d5d976 changed 11 files. <i>No additional rationale was recorded in the commit body.</i> Evidence E-HIS-D3E418A597

History boundary

- Git history contains 1923 commits; History recovered the most recent bounded 300. Older commits remain outside this projection unless identified by a recovered tag.

TRACEABILITY & EVIDENCE COVERAGE

How completely Bóveda reconstructed the project

Reconstruction confidence combines the evidence trail behind the main result with whether important applicable analytical areas could be reconstructed. It does not measure model quality, project quality, performance, compliance or governance.

Project reconstruction

Partial reconstruction

Bóveda could trace the main result to the data, model and evaluation sample behind it. However, 2 important applicable analytical areas could not be reconstructed, so the overall reconstruction remains partial.

- ✓ The recorded result can be identified.
- ✓ The result is connected to its recorded metric and value.
- ✓ The result is connected to a model and target.
- ✓ The evaluation context can be traced.
- ✓ The evaluation sample and main result can be traced together.
- ✓ Findings retain links to verification checks and Evidence Tokens.

Evidence coverage

Coverage not established

Project Context

Sufficient

Whether the project purpose, scope, methods, decisions and limitations are supported.

Traceability

Not assessed

Whether the main conclusions link to identifiable source material.

Reproducibility

Partial

Whether the main analytical result can be recreated from the available project material.

Source

Sufficient

Whether the source material needed to support the main claims was available for review.

INTERPRETATION BOUNDARY

Remaining limitations

What the client should not conclude from this report, based on the material boundaries represented in the current project record.

- 01 Excluding segments without historical crashes removes much of the road network, limiting conclusions about lower-history or previously crash-free segments.
- 02 The benchmark excludes segments without recorded crashes and requires several successful data joins, restricting interpretation beyond this selected subset.
- 03 It does not show performance across all city segments or that acting on risk scores prevents crashes.
- 04 Bóveda found the project data, but not enough information to identify the precise dataset snapshot behind

the reported result. A precise data version is needed to reproduce the same analysis from the same inputs.

05 Bóveda found the reported metric, but not a complete link to the calculation and score inputs that produced it. Without that calculation trail, Bóveda cannot fully verify how the reported value was obtained.

06 The available project record does not contain enough information to reconstruct the complete link between predictions and evaluation labels. Without that link, Bóveda cannot fully reproduce or independently verify the reported evaluation result.

07 Bóveda found an applicable classification result, but no linked diagnostic evidence showing false positives, false negatives or other behaviour behind the headline score. Without diagnostic evidence, a supervisor cannot see important error patterns that may be hidden by the headline metric.

08 Bóveda found missing-value handling in the focal analytical workflow, but no stage- and denominator-aware measurements showing how much data was missing. Without those measurements, a supervisor cannot assess the scale of missingness or how much the handling step affected the analytical data.

Absence of evidence is not evidence of absence. These statements are limited to the available project record reviewed by Bóveda.

Technical appendix

Detailed reconstruction retained for technical review. Internal identities and precise analytical terminology are intentionally confined to this layer.

Complete target definitions

TARGET	SOURCE FIELD	CONSTRUCTION	POSITIVE CLASS	INTERPRETATION	STATUS & EVIDENCE
Any crash on a segment in week 19 of 2017	–	Complete construction ancestry not established	–	Focal target reconstructed; technical construction links remain incomplete	OBSERVED E-3102C1ACD6

Target-construction ancestry

No supported entries were reconstructed for this table.

Full population and sample table

POPULATION	ROLE	COUNT	UNIT	PREDICATE	STATUS & EVIDENCE
Source rows	source	22,466	observations	—	OBSERVED E-B7DD9F89DA
Eligible population	eligible population	22,405	observations	Segments required a historical crash and successful joins to segment, traffic-recording and concern data before the benchmark split.	OBSERVED E-3102C1ACD6 · E-38DA02CA70
Labelled population	labelled population	3,309	observations	Segments required a historical crash and successful joins to segment, traffic-recording and concern data before the benchmark split.	DERIVED E-3102C1ACD6
Training population	training	2,283	observations	A random 70% training split supported five-fold randomized tuning; final ROC AUC was measured on the remaining 1,026 records.	OBSERVED E-3102C1ACD6 · E-E59BA1BCCB
Held-out evaluation population	evaluation	1,026	observations	A random 70% training split supported five-fold randomized tuning; final ROC AUC was measured on the remaining 1,026 records.	OBSERVED E-3102C1ACD6 · E-E59BA1BCCB
Actual scoring population in export code	actual scoring	22,405	observations	Rows supplied to a persisted prediction/export statement	OBSERVED E-3102C1ACD6 · E-E59BA1BCCB · E-E20AB3DB80

Complete model-comparison tables

Crash prediction benchmark

Results recorded for the same target and evaluation setup.

METRIC	RANDOM FOREST	LOGISTIC REGRESSION	XGBOOST
ROC AUC	0.51	0.507	0.736
Brier loss	0.083	0.215	0.149
F1	0.081	0.043	0.059

XGBoost records the strongest ROC AUC (0.7355), while Random Forest leads on Brier loss (0.0826). No single model leads across every available metric, and the rows should be interpreted separately because they measure different aspects of performance.

Evidence E-3102C1ACD6

Crash prediction benchmark

Results recorded for the same target and evaluation setup.

METRIC	RANDOM FOREST	LOGISTIC REGRESSION	XGBOOST
ROC AUC	0.477	0.507	0.736
Brier loss	0.082	0.215	0.149
F1	0.082	0.043	0.059

XGBoost records the strongest ROC AUC (0.7355), while Random Forest leads on Brier loss (0.0824). No single model leads across every available metric, and the rows should be interpreted separately because they measure different aspects of performance.

Evidence E-3102C1ACD6

Crash prediction benchmark

Results recorded for the same target and evaluation setup.

METRIC	XGBOOST	LOGISTIC REGRESSION
ROC AUC	0.736	0.636

XGBoost records the strongest value across all 1 comparable metrics shown for Crash prediction benchmark. The comparison is still row-specific: the values describe different evaluation measures and should not be compared vertically as though they shared one scale.

Evidence E-3102C1ACD6

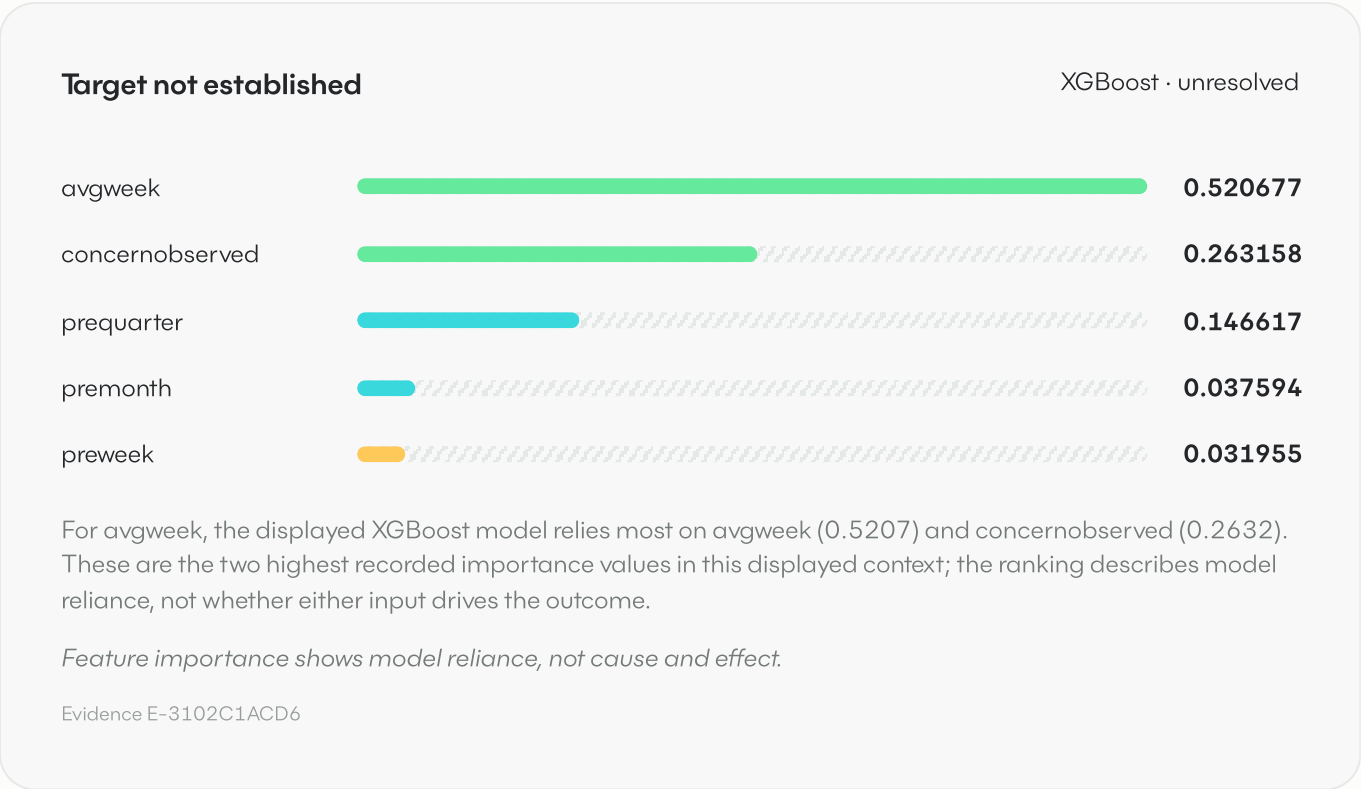
Prevalence and baseline context

No supported entries were reconstructed for this table.

Additional evaluation diagnostics

No supported entries were reconstructed for this table.

Full feature-evidence sets



Tuning and search stages

No supported entries were reconstructed for this table.

Model variants

TARGET	METHOD	ESTIMATOR	ROLES	PARAMETERS	SOURCE	EVIDENCE
Target not established	XGBoost	XGBClassifier	untuned	<code>**test.rundict['XG_base']['bp']</code>	notebooks/benchmark/crash_predict_benchmark.ipynb#cell-36	E-3102C1ACD6
Target not established	Logistic Regression	LogisticRegression	untuned	<code>**test.rundict['LR_base']['bp']</code>	notebooks/benchmark/crash_predict_benchmark.ipynb#cell-37	E-3102C1ACD6
Target not established	Logistic Regression	LogisticRegression	untuned	<code>**test.rundict['LR_base']['bp']</code>	notebooks/benchmark/crash_predict_benchmark.ipynb#cell-38	E-3102C1ACD6

Selection statements

Output relationships

TARGET	PRODUCER	OUTPUT	PATH	EXISTENCE	CONSISTENCY	EVIDENCE
Target not established	Unresolved	analytical data export	seg_with_risk_score_adj.csv	referenced not persisted	no structural mismatch detected	E-3102C1ACD6
Target not established	Unresolved	analytical data export	seg_with_risk_score.csv	referenced not persisted	no structural mismatch detected	E-E59BA1BCCB
Target not established	Unresolved	analytical data export	../data/processed/car_preds_weekly.csv	referenced not persisted	no structural mismatch detected	E-00147FE2AC
Target not established	Unresolved	analytical data export	../data/boston/processed/jam_diff.csv	referenced not persisted	no structural mismatch detected	E-E20AB3DB80
Target not established	Unresolved	analytical data export	crash_road_12_14_MA_unique.csv	referenced not persisted	no structural mismatch detected	E-3D83962C95
Target not established	Unresolved	analytical data export	intersection_crashes.csv	referenced not persisted	no structural mismatch detected	E-3D83962C95

Scoped limitations

No supported entries were reconstructed for this table.

Checks performed

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-CURRENT-PURPOSE-TRACE	Current purpose source	Not detected	implemented	Current purpose source is resolved to the inspected evidence.	E-B335630551
CHK-PROJECT-CONTEXT-COVERAGE	Material project context	Not detected	implemented	Material project context is resolved to the inspected evidence.	E-B335630551 · E-C998E295B4 · E-3102C1ACD6
CHK-SOURCE-INVENTORY-COVERAGE	Authorised source inventory	Not detected	implemented	Authorised source inventory is resolved to the inspected evidence.	E-B335630551 · E-3102C1ACD6 · E-B7DD9F89DA
CHK-TYPED-FOCAL-RESULT-ORIGIN	Typed focal-result origin	Not detected	implemented reconstruction native	The main result is connected to an evidence-backed target, model, evaluation context, metric and evaluation population.	E-3102C1ACD6
CHK-TARGET-CONSTRUCTION-ANCESTRY	Complete target construction ancestry	Not enough evidence	implemented reconstruction native	Bóveda reconstructed Any crash on a segment in week 19 of 2017, but not the complete sequence of source fields and transformations that created it.	E-3102C1ACD6
CHK-POPULATION-LINEAGE-COMPATIBILITY	Population lineage compatibility	Not detected	implemented graph native	Population roles, directed branches, units, partitions, and authorised count derivations are compatible.	E-3102C1ACD6 · E-38DA02CA70 · E-B7DD9F89DA · E-E20AB3DB80 · E-E59BA1BCCB
CHK-EVALUATION-INTENDED-POPULATION-ALIGNMENT	Evaluation versus intended prediction population	Not applicable	implemented graph native	No population-specific intended prediction or decision population is represented.	
CHK-EVALUATION-PREPROCESSOR-FIT-SCOPE	Evaluation preprocessing fit scope	Not enough evidence	implemented graph native	The project shows fitted data preparation, but the available evidence does not establish whether its parameters were learned from training data only or also from evaluation data.	E-784758FCB5

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented reconstruction native	ROC AUC uses a compatible probability based input.	E-3102C1ACD6
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented reconstruction native	ROC AUC uses a compatible probability based input.	E-3102C1ACD6
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented reconstruction native	ROC AUC uses a compatible probability based input.	E-3102C1ACD6
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not applicable	implemented graph native	No final binary class-decision diagnostic is represented.	
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Not enough evidence	implemented reconstruction native	The evaluation record describes model tuning, but does not establish the final search stage, its selected settings and the exact evaluated estimator needed for this comparison.	E-3102C1ACD6
CHK-OUTPUT-COLUMN-CONSISTENCY	Output column consistency	Not applicable	implemented graph native	No prediction output-production statement is represented.	
CHK-ANALYTICAL-ROLE-ALIGNMENT	Analytical model-role alignment	Not detected	implemented reconstruction native	The main-result and feature-producing model roles for Any crash on a segment in week 19 of 2017 both resolve to XGBoost.	E-3102C1ACD6

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-REPRODUCTION-INPUT-COVERAGE-V2	Result reproduction inputs	Not enough evidence	implemented graph native	One or more independent inputs needed to reproduce the result are missing in the available project evidence.	E-00147FE2AC · E-1BE4DCDB4E · E-2D321D0CCB · E-3102C1ACD6 · E-33DC416CAE · E-38DA02CA70 · E-3980CB24A2 · E-3D83962C95 · E-49694FAF5B · E-4D7C51B1EF · E-68417CB343 · E-784758FCB5 · E-B7DD9F89DA · E-C341F43CFC · E-C6EF786BDD · E-DD2C0EB6EA · E-E0D7C4E878 · E-E20AB3DB80 · E-E59BA1BCCB · E-EC0B19537E · E-F313830786
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Main result	Not detected	implemented analytical area native	Main result answers its defined supervisory question with connected evidence.	E-3102C1ACD6 · E-B7DD9F89DA
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Model comparison	Not detected	implemented analytical area native	Model comparison answers its defined supervisory question with connected evidence.	E-3102C1ACD6
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Feature / driver evidence	Not detected	implemented analytical area native	Feature / driver evidence answers its defined supervisory question with connected evidence.	E-3102C1ACD6
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Evaluation behaviour	Not enough evidence	implemented analytical area native	Bóveda found enough evidence to establish that evaluation behaviour is applicable, but not enough to answer the supervisory question that the section represents.	E-3102C1ACD6
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Population and sample lineage	Not detected	implemented analytical area native	Population and sample lineage answers its defined supervisory question with connected evidence.	E-3102C1ACD6 · E-B7DD9F89DA
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Missing data before preparation	Not enough evidence	implemented analytical area native	Bóveda found enough evidence to establish that missing data before preparation is applicable, but not enough to answer the supervisory question that the section represents.	E-3102C1ACD6

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-MATERIAL-EVIDENCE-ABSENCE	Material evidence absence	Condition found	implemented finding native	Material evidence absence prevents meaningful supervisory assessment of evaluation behaviour.	E-3102C1ACD6
CHK-MATERIAL-EVIDENCE-ABSENCE	Material evidence absence	Condition found	implemented finding native	Material evidence absence prevents meaningful supervisory assessment of missing data before preparation.	E-3102C1ACD6

Complete History projection

DATE	TYPE	EVENT	DESCRIPTION	ROLE	EVIDENCE
2016–2017	data period	Project data period reconstructed	Data through week 19 of 2017; concerns summarized for 2016	Material	E-3102C1ACD6
2017	evaluation	Primary evaluation recorded	Persisted project evidence records ROC AUC 0.73553583169 for Any crash on a segment in week 19 of 2017 using XGBoost classifier using five features.	Material	E-3102C1ACD6
7 Jun 2018	git tag	Release tag v1.1	Tag v1.1 identifies 26d8299c5c: Merge pull request #117 from Data4Democracy/docker	Material	E-HIS-CD26C02968
12 Aug 2018	git tag	Release tag v1.2	Tag v1.2 identifies c79a562eea: Merge pull request #160 from Data4Democracy/fix_bug	Material	E-HIS-C55CAB55D7
20 Mar 2019	git tag	Release tag v2.0	Tag v2.0 identifies 37de9d9855: Update readme to new slack channel name	Material	E-HIS-63857821EE
14 Aug 2019	git commit	Make canon dataset updates	Commit 17234a1ca1 changed 3 files.	Material	E-HIS-FCF562DDCC
28 Aug 2019	git commit	added split column capability to model	Commit 56284d0f20 changed 2 files.	Material	E-HIS-5E472FA611
9 Sept 2019	git commit	Update make_preds_viz + pipeline to generate postprocessed files for split_columns	Commit 434780a182 changed 2 files.	Material	E-HIS-A19507381C
8 Oct 2019	git commit	Remove risk_map from pipeline	Commit 6b22e7085c changed 1 file.	Material	E-HIS-574EA43B97
20 Dec 2019	git commit	Added new city config for NYC. Ran pipeline and tested the web app, works but does not show modes yet.	Commit 6edc5c8c92 changed 1 file.	Material	E-HIS-EFAA393053
25 Jun 2020	git commit	fix to model use appropriate featureset	Commit 9a5203a5ec changed 1 file.	Material	E-HIS-6E325D0345
30 Dec 2021	git commit	adaptation for multi feature doc fixes and segment format fix	Commit 9fb8d5d976 changed 11 files.	Material	E-HIS-D3E418A597

DATE	TYPE	EVENT	DESCRIPTION	ROLE	EVIDENCE
26 Aug 2026, 11:55 UTC	boveda audit	Bóveda audit completed	Bóveda completed audit AUD-8F3965C55C and stored the current canonical Project Record. One reconstruction attempt was recorded.	Material	

The PDF retains material History rows. 294 supporting project changes remain available in the complete HTML report.

Technical provenance

ANALYTICAL GRAPH SCHEMA boveda-analytical-object-graph-0.13.2	DERIVATION "deterministic_persisted_evidence_v0.13.2"
COMPLETENESS <pre>{"targets":"unavailable","populations":"available","model_runs":"available","searches":"unavailable","metrics":"unavailable","diagnostics":"unavailable","prevalence":"unavailable","outputs":"available","environments":"available","evidence_termination":"complete"}</pre>	PROVIDER CALLS USED BY THIS ANALYTICAL PROJECTION 0

Evidence appendix

A stable index linking material report claims to source evidence retained in the Project Record.

Evidence Index

EVIDENCE TOKEN	SOURCE	SOURCE LOCATOR	EPISTEMIC STATUS	SHORT DESCRIPTION
E-00147FE2AC	notebooks/python2/crash_predict_car.ipynb	notebooks/python2/crash_predict_car.ipynb	OBSERVED	[CODE FOR OUTPUT] class CAR(pm.distributions.distribution.Continuous): def __init__(self, alpha, adjacency, *args, **kwargs): if not isinstance(adjacency, np.ndarray): raise ValueError("A...
E-1BE4DCDB4E	conda-win-64.lock	conda-win-64.lock	OBSERVED	# platform: win-64 # env.hash: f32645531ce8ff0bdf703d6ad1beea7306227132570eda0e6be08cd8d 9f071d9 @EXPLICIT https://repo.anaconda.com/pkgs/main/win-64/_py- xgboost-mutex-2.0-cpu_0.conda#b783...
E-2D321D0CCB	notebooks/python2/tutorial.ipynb	notebooks/python2/tutorial.ipynb	OBSERVED	[CODE FOR OUTPUT] items = json.load(open(PROCESSED_DATA_FP + 'crash_joined.json')) crash_list, crashes_by_id = util.group_json_by_location(items) print str(len(crash_list)) + " crashes fo...
E-3102C1ACD6	notebooks/benchmark/crash_predict_benchmark.ipynb	notebooks/benchmark/crash_predict_benchmark.ipynb	OBSERVED	[CODE FOR OUTPUT] import numpy as np import pandas as pd import sklearn.ensemble as ske import sklearn.svm as svm import sklearn.linear_model as skl import xgboost as xgb import seaborn a...
E-33DC416CAE	environment_linux.yml	environment_linux.yml	OBSERVED	name: crash-model channels: - conda-forge - defaults dependencies: - _libgcc_mutex=0.1=conda_forge - _openmp_mutex=4.5=1_gnu - _py- xgboost-mutex=2.0=cpu_0 - argon2-cffi=20.1.0=py36h8f6f2f...
E-38DA02CA70	notebooks/route_level_risk_distribution.ipynb	notebooks/route_level_risk_distribution.ipynb	OBSERVED	[CODE FOR OUTPUT] # on average, how different are the preds print(segments_pred.shape) all_preds = segments_pred.merge(p_segments_pred[['segment_id', 'prediction']], on='segment_id' pri...
E-3980CB24A2	conda-linux-64.lock	conda-linux-64.lock	OBSERVED	# platform: linux-64 # env.hash: 3dbe3d001c92088bd21fd687fd427fd8bbe8436cc64ffd2053ecc75425 b3a7c4 @EXPLICIT https://conda.anaconda.org/conda-forge/linux- 64/_libgcc_mutex-0.1-conda_forge.t...
E-3D83962C95	notebooks/python2/data_explore_12_14_crashes_road_details.ipynb	notebooks/python2/data_explore_12_14_crashes_road_details.ipynb	OBSERVED	[CODE FOR OUTPUT] # read road data from csv to create a dataframe and convert column names to lowercase dbf_roads_data = Dbf5('boston_road_massdot2015_MA.dbf', codec='utf-8') road_data = d...
E-49694FAF5B	notebooks/python2/TMC_analysis.ipynb	notebooks/python2/TMC_analysis.ipynb	OBSERVED	[CODE FOR OUTPUT] PROCESSED_DATA_FP = '../osm-data/processed/' inters = {} with open(PROCESSED_DATA_FP + 'inters_data.json') as f: data = json.load(f) for key, value in data.iteritems():...
E-4D7C51B1EF	requirements.txt	requirements.txt	OBSERVED	codecov==2.1.13 fiona==1.9.4 flask==3.0.2 folium==0.16.0 geopandas==0.14.3 geocoder==1.38.1 geojson==3.1.0 matplotlib==3.8.3 numpy==1.26.4 openpyxl==3.1.2 osmnx==1.9.1 pandas==2.2.1 pypro...

EVIDENCE TOKEN	SOURCE	SOURCE LOCATOR	EPISTEMIC STATUS	SHORT DESCRIPTION
E-68417CB343	conda-osx-64.lock	conda-osx-64.lock	OBSERVED	# platform: osx-64 # env.hash: 2028719b58bd46aff45f497cc1abfe1f0b6a59e01a57d04e4fb4141d93 4c7014 @EXPLICIT https://conda.anaconda.org/conda-forge/osx- 64/_py-xgboost-mutex-2.0-cpu_0.tarbz2...
E-784758FCB5	notebooks/clustering_segment_lev el.ipynb	notebooks/clustering_segment_lev el.ipynb	OBSERVED	[CODE FOR OUTPUT] df.input = df.minimal_columns.fillna(0) #df.input = test_dummies.fillna(0) pca = PCA(n_components=4)#,random_state=12345) # make pipeline pipe = make_pipeline(StandardSc...
E-B335630551	README.md	README.md	OBSERVED	Crash Modeling ===== Outline: ----- -- - Project Overview - Data Sources and Modelling - Setting up - Contributing - Connect with us - Project Organization Pr...
E-B7DD9F89DA	notebooks/vision_zero_analysis.ipy nb	notebooks/vision_zero_analysis.ipy nb	OBSERVED	[CODE FOR OUTPUT] import os sys.path.append('../src') BASE_DIR = os.path.dirname(os.getcwd()) print(BASE_DIR) print(os.getcwd()) from data import util from data import analysis_util [PERS...
E-C341F43CFC	src/showcase/Dockerfile	src/showcase/Dockerfile	OBSERVED	FROM python:alpine3.7 RUN pip install flask COPY .
E-C6EF786BDD	environment_mac.yml	environment_mac.yml	OBSERVED	name: crash-model channels: - conda-forge dependencies: - _py-xgboost- mutex=2.0 - affine=2.4.0 - attrs=23.2.0 - aws-c-auth=0.7.16 - aws-c- cal=0.6.10 - aws-c-common=0.9.13 - aws-c-compress...
E-C998E295B4	src/visualization/README.md	src/visualization/README.md	OBSERVED	# Crash model visualization This directory contains the code relevant to the visualization efforts for this project.
E-DD2C0EB6EA	Dockerfile	Dockerfile	OBSERVED	FROM continuumio/miniconda3 # update RUN conda update -n base conda # Set package installer as non-interactive ENV DEBIAN_FRONTEND noninteractive # Set a terminal type ENV TERM xterm-256c...
E-E0D7C4E878	environment_pc.yml	environment_pc.yml	OBSERVED	name: crash-model channels: - conda-forge - defaults dependencies: - _py-xgboost-mutex=2.0=cpu_0 - astroid=2.4.2=py36h9f0ad1d_0 - atomicwrites=1.4.0=pyh9f0ad1d_0 - attrs=19.3.0=py_0 - bac...
E-E1D5879CE3	src/README.md	src/README.md	OBSERVED	# Crash model pipeline This directory contains the code for taking the data from the csv crash files, standardizing the crash data, generating the feature set, training a model to predict...
E-E20AB3DB80	notebooks/intervention_effect.ipyn b	notebooks/intervention_effect.ipyn b	OBSERVED	[CODE FOR OUTPUT] changed = preds_df.diff(axis=1).iloc[:,1]!=0 p = preds_df[changed].plot(kind='kde') p.set_xlabel('risk score') p.legend(labels=['As-is','20 speed limit']) [PERSISTED OUT...

EVIDENCE TOKEN	SOURCE	SOURCE LOCATOR	EPISTEMIC STATUS	SHORT DESCRIPTION
E-E59BA1BCCB	notebooks/benchmark/crash_predict_rnn.ipynb	notebooks/benchmark/crash_predict_rnn.ipynb	OBSERVED	[CODE FOR OUTPUT] # split into train and test sets train_size = int(len(data_model) * 0.67) test_size = len(data_model) - train_size #train, test = data_model[0:train_size:], data_model[...
E-EC0B19537E	notebooks/python2/open_street_map.ipynb	notebooks/python2/open_street_map.ipynb	OBSERVED	[CODE FOR OUTPUT] node_counts = {} dup_nodes = {} # Any node that's shared by more than one way is an intersection non_named_count = 0 way_lines = [] unnamed_lines = [] service_lines = []...
E-F313830786	notebooks/general_analysis.ipynb	notebooks/general_analysis.ipynb	OBSERVED	[CODE FOR OUTPUT] items = json.load(open(PROCESSED_DATA_FP + 'crash_joined.json')) crash_list, crashes_by_id = util.group_json_by_location(items) print(str(len(crash_list)) + " crashes fo...
E-HIS-574EA43B97	git:6b22e7085cfbb30e9a84e5338898141377d0a01c	git:6b22e7085cfbb30e9a84e5338898141377d0a01c	OBSERVED	Commit: 6b22e7085cfbb30e9a84e5338898141377d0a01c Authored: 2019-10-08T14:31:23-04:00 Subject: Remove risk_map from pipeline Body: Not recorded Changed files (1): M src/pipeline.py
E-HIS-5E472FA611	git:56284d0f20d072ffd1cc2b05efd2a400f9d1655e	git:56284d0f20d072ffd1cc2b05efd2a400f9d1655e	OBSERVED	Commit: 56284d0f20d072ffd1cc2b05efd2a400f9d1655e Authored: 2019-08-28T17:11:50-04:00 Subject: added split column capability to model Body: Not recorded Changed files (2): M src/models/tra...
E-HIS-63857821EE	git:tag/v2.0	git:tag/v2.0	OBSERVED	Tag: v2.0 Commit: 37de9d9855c1387a66cea4ce777052cd032d4ee7 Created: 2019-03-20T17:52:20-04:00 Subject: Update readme to new slack channel name
E-HIS-6E325D0345	git:9a5203a5ec77d1d53afb361555398aa122db8a1e	git:9a5203a5ec77d1d53afb361555398aa122db8a1e	OBSERVED	Commit: 9a5203a5ec77d1d53afb361555398aa122db8a1e Authored: 2020-06-25T17:05:21-04:00 Subject: fix to model use appropriate featureset Body: Not recorded Changed files (1): M src/models/tr...
E-HIS-A19507381C	git:434780a182b1fa7d1908454423b932bb8143062c	git:434780a182b1fa7d1908454423b932bb8143062c	OBSERVED	Commit: 434780a182b1fa7d1908454423b932bb8143062c Authored: 2019-09-08T18:24:54-04:00 Subject: Update make_preds_viz + pipeline to generate postprocessed files for split_columns Body: Not...
E-HIS-C55CAB55D7	git:tag/v1.2	git:tag/v1.2	OBSERVED	Tag: v1.2 Commit: c79a562eea13f1bc8532c0d40a04113ff48c7ab0 Created: 2018-08-12T11:32:26-04:00 Subject: Merge pull request #160 from Data4Democracy/fix_bug
E-HIS-CD26C02968	git:tag/v1.1	git:tag/v1.1	OBSERVED	Tag: v1.1 Commit: 26d8299c5c203a5437ca2b2db7a076d7cc3b6f69 Created: 2018-06-06T20:40:19-04:00 Subject: Merge pull request #117 from Data4Democracy/docker
E-HIS-D3E418A597	git:9fb8d5d9776e594b3c339048434a0aa0edbaaed45	git:9fb8d5d9776e594b3c339048434a0aa0edbaaed45	OBSERVED	Commit: 9fb8d5d9776e594b3c339048434a0aa0edbaaed45 Authored: 2021-12-30T10:51:10-05:00 Subject: adaptation for multi feature doc fixes and segment format fix Body: Not recorded Changed file...

EVIDENCE TOKEN	SOURCE	SOURCE LOCATOR	EPISTEMIC STATUS	SHORT DESCRIPTION
E-HIS-EFAA393053	git:6edc5c8c92858d614450d10f12407e2b25bc9950	git:6edc5c8c92858d614450d10f12407e2b25bc9950	OBSERVED	Commit: 6edc5c8c92858d614450d10f12407e2b25bc9950 Authored: 2019-12-19T19:20:13-05:00 Subject: Added new city config for NYC.
E-HIS-FCF562DDCC	git:17234a1ca127727b27df67e5c8d143144ab4f4e1	git:17234a1ca127727b27df67e5c8d143144ab4f4e1	OBSERVED	Commit: 17234a1ca127727b27df67e5c8d143144ab4f4e1 Authored: 2019-08-14T13:00:13-04:00 Subject: Make canon dataset updates Body: Not recorded Changed files (3): M src/features/make_canon_da...

The PDF index includes evidence linked to material report content. 305 additional supporting History entries remain indexed and expandable in the complete HTML report.

Raw evidence excerpts

Raw excerpts are separated from the client narrative. Browser readers may expand every retained excerpt, and longer material is clearly marked as partial in print. The PDF includes bounded excerpts for evidence linked to material report content; supporting source-control History remains complete in the self-contained HTML report.

E-00147FE2AC · notebooks/python2/crash_predict_car.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

[CODE FOR OUTPUT]

```
class CAR(pm.distributions.distribution.Continuous):
    def __init__(self, alpha, adjacency, *args, **kwargs):
        if not isinstance(adjacency, np.ndarray):
            raise ValueError("Adjacency matrix is not an ndarray.")
        n, m = adjacency.shape
        if n != m or np.any(adjacency != adjacency.T):
            raise ValueError('Adjacency matrix must be symmetric.')
        if 'shape' in kwargs and kwargs['shape'] == n:
            raise ValueError('Invalid shape: Must match matrix

[Middle of retained excerpt omitted from the print edition.]

E]
# output predicted probabilities
pd.DataFrame(zip(data_model['segment_id'].values, pred_y),
              columns=['segment_id', 'pred']).to_csv('../data/processed/car_preds.csv')
```

E-1BE4DCDB4E · conda-win-64.lock

Epistemic status: OBSERVED · **Kind:** environment or dependency manifest

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
# platform: win-64
# env_hash: f32645531ce8ff0bdf703d6ad1beea7306227132570eda0e6be08cd8d9f071d9
@EXPLICIT
https://repo.anaconda.com/pkgs/main/win-64/_py-xgboost-mutex-2.0-
cpu_0.conda#b78304ae04e805fdef3625f7d1fe7e86
https://repo.anaconda.com/pkgs/main/win-64/ca-certificates-2021.1.19-
haa95532_0.conda#be6f2398b9ef7ee609fb9bf8505a31e7
https://conda.anaconda.org/conda-forge/win-64/intel-openmp-2020.3-
h57928b3_311.tar.bz2#361a51c98a0cb06415bd3f9d3bbb5778
https://conda.anaconda.org/conda-forge/win-64/msys2-conda-epoch-2
```

[Middle of retained excerpt omitted from the print edition.]

```
gets-7.6.3-pyhd3deb0d_0.tar.bz2#536a9ed6d9e740f2b83d1a3c388e4388
https://conda.anaconda.org/conda-forge/win-64/jupyter-1.0.0-
py36ha15d459_6.tar.bz2#85320b332c77dab3063669b7d741a797
```

E-2D321D0CCB · notebooks/python2/tutorial.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
[CODE FOR OUTPUT]
items = json.load(open(PROCESSED_DATA_FP + 'crash_joined.json'))
crash_list, crashes_by_id = util.group_json_by_location(items)
print str(len(crash_list)) + " crashes found"

# Crash_list is a list of dicts about each individual crash
print crash_list[0]

# Crashes by id is a dict where the key is the segment id. It contains the count of crashes at that
intersection
# Can also contain other information about this location, if called with arg otherfields. In this case,
we're
# Looking at time
sor
```

[Middle of retained excerpt omitted from the print edition.]

```
FP = '../osm-data/processed/'
```

```
[CODE]
import json
import os
BASE_DIR = os.path.dirname(os.getcwd())
os.chdir(BASE_DIR + '/src/data/')
import util
os.chdir(BASE_DIR + '/notebooks/')

```


E-3102C1ACD6 · notebooks/benchmark/crash_predict_benchmark.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
[CODE FOR OUTPUT]
```

```
import numpy as np
import pandas as pd
import sklearn.ensemble as ske
import sklearn.svm as svm
import sklearn.linear_model as skl
import xgboost as xgb
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn import metrics
from sklearn.model_selection import RandomizedSearchCV, KFold, StratifiedKFold, GroupKFold,
GroupShuffleSplit
from sklearn.calibration import CalibratedClassifierCV
from sklearn.preprocessing import StandardScaler
```

```
class Indata():
    scoring = None
    data = None
```

```
[Middle of retained excerpt omitted from the print edition.]
```

```
'crash', w, 2016)
    data = crash_lags.merge(data_segs, left_on='segment_id', right_on='segment_id')
    adj_lags = get_adj_crash_lags(w, 2016)
    data = data.merge(
```

E-33DC416CAE · environment.linux.yml

Epistemic status: OBSERVED · **Kind:** environment or dependency manifest

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
name: crash-model
channels:
  - conda-forge
  - defaults
dependencies:
  - _libgcc_mutex=0.1=conda_forge
  - _openmp_mutex=4.5=1_gnu
  - _py-xgboost-mutex=2.0=cpu_0
  - argon2-cffi=20.1.0=py36h8f6f2f9_2
  - astroid=2.4.2=py36h9f0ad1d_1
  - async_generator=1.10=py_0
  - attrs=20.3.0=pyhd3deb0d_0
  - backcall=0.2.0=pyh9f0ad1d_0
  - backports=1.0=py36_1
  - backports.functools_lru_cache=1.6.1=py_0
  - bleach=3.2.3=pyh44b312d_0
  - boost-cpp=1.70.0=ha2d47e9_1
  - branca=0.4.2=pyhd8ed1ab_0
  - brotlipy=0.7.0=py36he6145b
```

[Middle of retained excerpt omitted from the print edition.]

```
- xz=5.2.5=h516909a_1
  - yaml=0.2.5=h516909a_0
  - zeromq=4.3.3=h58526e2_3
  - zipp=3.4.0=py_0
  - zlib=1.2.11=h516909a_1010
  - zstd=1.4.8=ha95c52a_1
prefix: [local path removed]
```

E-38DA02CA70 · notebooks/route_level_risk_distribute.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
[CODE FOR OUTPUT]
# on average, how different are the preds
print(segments_pred.shape)
all_preds = segments_pred.merge(
    p_segments_pred[['segment_id', 'prediction']],
    on='segment_id')
print(all_preds.shape)
all_preds = all_preds.merge(
    b_segments_pred[['segment_id', 'prediction']],
    on='segment_id')
print(all_preds.shape)
all_preds.rename(columns={
    'prediction_x': 'v_pred',
    'prediction_y': 'p_pred',
    'prediction': 'b_pred'},
    inplace=True)
(all_preds.filter(regex='pred').max(axis=1) - all_

[Middle of retained excerpt omitted from the print edition.]

red_cols
# calculate risk per m
risk_per_m = r_df[pred_cols].apply(lambda x: x/r_df['length'])
risk_per_m['route_id'] = r_df['route_id']

[CODE]
city_map.save('school_routes.html')
```

E-3980CB24A2 · conda-linux-64.lock

Epistemic status: OBSERVED · **Kind:** environment or dependency manifest

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
# platform: linux-64
# env_hash: 3dbe3d001c92088bd21fd687fd427fd8bbe8436cc64ffd2053ecc75425b3a7c4
@EXPLICIT
https://conda.anaconda.org/conda-forge/linux-64/_libgcc_mutex-0.1-
conda_forge.tar.bz2#d7c89558ba9fa0495403155b64376d81
https://conda.anaconda.org/conda-forge/linux-64/_py-xgboost-mutex-2.0-
cpu_0.tar.bz2#23b8f98a355030331f40d0245492f715
https://repo.anaconda.com/pkgs/main/linux-64/ca-certificates-2021.1.19-
h06a4308_0.conda#c73203541f779cf79629e9dbe62f5e0b
https://conda.anaconda.org/conda-forge/linux-64/ld_impl

[Middle of retained excerpt omitted from the print edition.]

ts-7.6.3-pyhd3deb0d_0.tar.bz2#536a9ed6d9e740f2b83d1a3c388e4388
https://conda.anaconda.org/conda-forge/linux-64/jupyter-1.0.0-
py36h5fab9bb_6.tar.bz2#229db303c1997c7005d240e4df7208b7
```

E-3D83962C95 · notebooks/python2/data_explore_12_14_crashes_road_details.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

[CODE FOR OUTPUT]

```
# read road data from csv to create a dataframe and convert column names to lowercase
dbf_roads_data = Dbf5('boston_road_massdot2015_MA.dbf', codec='utf-8')
road_data = dbf_roads_data.to_dataframe()
road_data.columns = road_data.columns.str.lower()
print road_data.columns
```

[PERSISTED OUTPUT]

```
Index([u'objectid', u'route_id', u'from_measu', u'to_measure', u'from_date',
       u'to_date', u'control', u'facility', u'f_class', u'urban_area',
       u'urban_type', u'f_f_class', u'jurisdictn', u'hwy_dist'])
```

[Middle of retained excerpt omitted from the print edition.]

```
oad_data.drop('strsegfid', axis=1, inplace=True)
crash_road_data.drop('mpo', axis=1, inplace=True)
```

```
# save merged dataset
crash_road_data.to_csv('crash_road_12_14_MA_unique.csv')
```

E-49694FAF5B · notebooks/python2/TMC_analysis.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

[CODE FOR OUTPUT]

```
PROCESSED_DATA_FP = '../osm-data/processed/'
```

```
inters = {}
with open(PROCESSED_DATA_FP + 'inters_data.json') as f:
    data = json.load(f)
    for key, value in data.iteritems():
        inters[str(key)] = value[0]
```

```
with open(PROCESSED_DATA_FP + 'tmc_summary.json') as f:
    addresses = json.load(f)
```

```
print addresses[0]
```

[PERSISTED OUTPUT]

```
{u'Right': 563, u'crash_count': 1, u'Longitude': u'-71.0631259', u'Filename': u'7436_2407_BUNKER-HILL-ST,-MYSTIC-ST,-SCHOOL-ST_NA_NA_CHARLESTOWN_11-HOURS_NA_0
```

[Middle of retained excerpt omitted from the print edition.]

```
fill_color='green', fill=True, fill_opacity=.7, color='grey',
radius=6.0).add_to(points)
```

```
display(points)
```

[CODE]

```
compare_crashes()
#plot_tmcs(addresses)
```

E-4D7C51B1EF · requirements.txt

Epistemic status: OBSERVED · **Kind:** environment or dependency manifest

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
codecov==2.1.13
fiona==1.9.4
flask==3.0.2
folium==0.16.0
geopandas==0.14.3
geocoder==1.38.1
geojson==3.1.0
matplotlib==3.8.3
numpy==1.26.4
openpyxl==3.1.2
osmnx==1.9.1
pandas==2.2.1
pyproj==3.6.1
pytest==8.0.2
pyyaml==6.0.1
pytest-cov==4.1.0
rtree==1.2.0
ruamel.yaml==0.17.33
scikit-learn==1.4.1.post1
shapely==2.0.3
tzlocal==5.2
xlrd==2.0.1
xgboost==2.0.3
tox==4.14.1
jsonschema==4.21.1
pylint==3.1.0
```

E-68417CB343 · conda-osx-64.lock

Epistemic status: OBSERVED · **Kind:** environment or dependency manifest

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
# platform: osx-64
# env_hash: 2028719b58bd46aff45f497cc1abfe1f0b6a59e01a57d04e4fb4141d934c7014
@EXPLICIT
https://conda.anaconda.org/conda-forge/osx-64/_py-xgboost-mutex-2.0-
cpu_0.tar.bz2#96dd44c2471a638fb9cdc40aa2573d1b
https://conda.anaconda.org/conda-forge/osx-64/bzip2-1.0.8-
hc929b4f_4.tar.bz2#521b82c700a64bac7d8da27539422e91
https://repo.anaconda.com/pkgs/main/osx-64/ca-certificates-2021.1.19-
hecd8cb5_0.conda#bd9c3c4901fe6a6e8563b104e983aac2
https://repo.anaconda.com/pkgs/main/osx-64/freexl-1.0.6-h9ed2024_0.con
```

[Middle of retained excerpt omitted from the print edition.]

```
gets-7.6.3-pyhd3deb0d_0.tar.bz2#536a9ed6d9e740f2b83d1a3c388e4388
https://conda.anaconda.org/conda-forge/osx-64/jupyter-1.0.0-
py36h79c6626_6.tar.bz2#4de1206a171612d881ade0758a4f31c8
```

E-784758FCB5 · notebooks/clustering_segment_level.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

[CODE FOR OUTPUT]

```
df_input = df_minimal_columns.fillna(0)
#df_input = test_dummies.fillna(0)
pca = PCA(n_components=4)#, random_state=12345)
# make pipeline
pipe = make_pipeline(StandardScaler(), pca)
pipe.fit(df_input)
pca_result = pipe.fit_transform(df_input)
```

[PERSISTED OUTPUT]

```
[local path removed] DataConversionWarning: Data with input dtype int64, float64 were all converted to float64 by StandardScaler.
```

```
    return self.partial_fit(X, y)
```

```
[local path removed] DataConversionWarning: Data with input dtype int64, flo
```

[Middle of retained excerpt omitted from the print edition.]

```
sub_cols = df.filter(regex=x).columns.tolist()
    if len(sub_cols)>2:
        sub_cols = sub_cols[1:]
    else:
        sub_cols = [sub_cols[0]]
    col_w_factors.extend(sub_cols)
```

E-B335630551 · README.md

Epistemic status: OBSERVED · **Kind:** documentation

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

Crash Modeling

=====

Outline:

- Project Overview
- Data Sources and Modelling
- Setting up
- Contributing
- Connect with us
- Project Organization

Project Overview

****Motivation****

This project was originally begun as a collaboration between Data4Democracy and the City of Boston.

On Jan 25th, 2017, [9 pedestrians were hit in Boston by vehicles]
(http://www.bostonherald.com/news/local_coverage/2017/01/battle_for_safer_streets_nine_pedestrians_hit_

[Middle of retained excerpt omitted from the print edition.]

based on the [cookiecutter data science project template](https://drivendata.github.io/cookiecutter-data-science/). #cookiecutterdatascience</small></p>

E-B7DD9F89DA · notebooks/vision_zero_analysis.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record. Source collection bounded this excerpt; the source may contain additional content. The HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

[CODE FOR OUTPUT]

```
import os
sys.path.append('../src')
BASE_DIR = os.path.dirname(os.getcwd())
print(BASE_DIR)
```

```
print(os.getcwd())
from data import util
from data import analysis_util
```

[PERSISTED OUTPUT]

```
[local path removed]
[local path removed]
```

[DETERMINISTIC POPULATION PARTITION]

Persisted output reports 7,790 intersections and 14,676 non-intersections as complementary categories in one population of segments.

Derived population size: 22,466 segment records.

Operation: 7,790 + 14,676 = 22,466.

Boundary: this es

[Middle of retained excerpt omitted from the print edition.]

	23	
20		14
21		19
22		33
23		

E-C341F43CFC · src/showcase/Dockerfile

Epistemic status: OBSERVED · **Kind:** environment or dependency manifest

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
FROM python:alpine3.7
RUN pip install flask
COPY . /app
WORKDIR /app
EXPOSE 5000
CMD ["python", "app.py"]
```

E-C6EF786BDD · environment_mac.yml

Epistemic status: OBSERVED · **Kind:** environment or dependency manifest

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
name: crash-model
channels:
  - conda-forge
dependencies:
  - _py-xgboost-mutex=2.0
  - affine=2.4.0
  - attrs=23.2.0
  - aws-c-auth=0.7.16
  - aws-c-cal=0.6.10
  - aws-c-common=0.9.13
  - aws-c-compression=0.2.18
  - aws-c-event-stream=0.4.2
  - aws-c-http=0.8.1
  - aws-c-io=0.14.5
  - aws-c-mqtt=0.10.2
  - aws-c-s3=0.5.2
  - aws-c-sdkutils=0.1.15
  - aws-checksums=0.1.18
  - aws-crt-cpp=0.26.2
  - aws-sdk-cpp=1.11.267
  - azure-core-cpp=1.11.1
  - azure-storage-blobs-cpp=12.10.0
  - azure-storage-common-cpp=12.5.
```

[Middle of retained excerpt omitted from the print edition.]

```
- xlrld=2.0.1
- xorg-libxau=1.0.11
- xorg-libxdmcp=1.1.3
- xyzservices=2023.10.1
- xz=5.2.6
- yaml=0.2.5
- zipp=3.17.0
- zlib=1.2.13
- zstd=1.5.5
- pylint==3.1.0
```

E-C998E295B4 · src/visualization/README.md

Epistemic status: OBSERVED · **Kind:** documentation

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

Crash model visualization

This directory contains the code relevant to the visualization efforts for this project.

Visualization Products

_risk_map.py_ – This script can be used to plot predictions generated from multiple models on a single Leaflet map of Boston. It color-codes each segment based on the magnitude of the predicted risk.

To run this script, you need the following inputs:

- inter_and_non_int.shp (created in create_segments.py)
- csv files of predictions (each row should have 1 prediction per

[Middle of retained excerpt omitted from the print edition.]

ojson and car_preds_named.json

- to run historical_crash_map.py to generate the data needed for the bar graph. Place the generated csv file in the same directory as the html file.

E-DD2C0EB6EA · Dockerfile

Epistemic status: OBSERVED · **Kind:** environment or dependency manifest

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

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```
FROM continuumio/miniconda3
# update
RUN conda update -n base conda

# Set package installer as non-interactive
ENV DEBIAN_FRONTEND noninteractive

# Set a terminal type
ENV TERM xterm-256color

WORKDIR /app

# Install packages
RUN apt-get update -qq && apt-get install -y --no-install-recommends \
    # apache for serving the visualisation
    apache2 \
    # easier management of services via supervisor
    supervisor \
    # base anaconda image seems to lack libgl support required for our virtual environment
    libgl1-mesa-glx \
    #

[Middle of retained excerpt omitted from the print edition.]

.bashrc

# this startup script runs supervisor in foreground (which in turn starts apache) to keep container
running
CMD ["/start.sh"]

# Make the apache port available
EXPOSE 8080
```

E-E0D7C4E878 · environment_pc.yml

Epistemic status: OBSERVED · **Kind:** environment or dependency manifest

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Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
name: crash-model
channels:
  - conda-forge
  - defaults
dependencies:
  - _py-xgboost-mutex=2.0=cpu_0
  - astroid=2.4.2=py36h9f0ad1d_0
  - atomicwrites=1.4.0=pyh9f0ad1d_0
  - attrs=19.3.0=py_0
  - backcall=0.2.0=pyh9f0ad1d_0
  - blas=1.0=mkl
  - bleach=3.1.5=pyh9f0ad1d_0
  - boost-cpp=1.72.0=h2ba7cf6_1
  - branca=0.4.1=py_0
  - brotlipy=0.7.0=py36h779f372_1000
  - bzip2=1.0.8=hfa6e2cd_2
  - ca-certificates=2020.6.20=hecda079_0
  - certifi=2020.6.20=py36h9f0ad1d_0
  - cffi=1.14.0=py36ha419a9e_0
```

[Middle of retained excerpt omitted from the print edition.]

```
.2=h6538335_2
  - zipp=3.1.0=py_0
  - zlib=1.2.11=h2fa13f4_1006
  - zstd=1.4.4=h9f78265_3
  - pip:
    - pyqt5-sip==4.19.18
    - pyqtchart==5.12
    - pyqtwebengine==5.12.1
```

E-E1D5879CE3 · src/README.md

Epistemic status: OBSERVED · **Kind:** documentation

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

Crash model pipeline

This directory contains the code for taking the data from the csv crash files, standardizing the crash data, generating the feature set, training a model to predict risk on segments, and visualizing the results.

Module dependencies

If using conda, you can get all the dependencies using the [environment_linux.yml] (https://github.com/Data4Democracy/crash-model/blob/master/environment_linux.yml), [environment_mac.yml] (https://github.com/Data4Democracy/crash-model/blob/master/environment_mac.yml)

[Middle of retained excerpt omitted from the print edition.]

pull request to get them to run), the file will appear in the results from the job, under Artifacts. You can download that and save it to the appropriate environment_<os>.yml file

E-E20AB3DB80 · notebooks/intervention_effect.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
[CODE FOR OUTPUT]
changed = preds_df.diff(axis=1).iloc[:,1]!=0
p = preds_df[changed].plot(kind='kde')
p.set_xlabel('risk score')
p.legend(labels=['As-is', '20 speed limit'])
```

```
[PERSISTED OUTPUT]
<matplotlib.legend.Legend at 0x10ee30240>
```

```
[PERSISTED OUTPUT]
<Figure size 432x288 with 1 Axes>
```

```
[CODE FOR OUTPUT]
# change no jams to some jams
jam_feats = [x for x in features if x.find('jam')>-1]
jam_df = df.copy(deep=True)
# has no jams
no_jams = jam_df.jam0==1
jam_df.loc[no_jams, 'jam0'] = 0
jam_df.loc[no_jams, 'jam1'] =
```

[Middle of retained excerpt omitted from the print edition.]

```
id_preds = id_preds.merge(streets, left_on='segment_id', right_on='id')
id_preds['diff'] = id_preds.after - id_preds.before
id_preds.to_csv('../data/boston/processed/jam_diff.csv')
```

E-E59BA1BCCB · notebooks/benchmark/crash_predict_rnn.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

[CODE FOR OUTPUT]

```
# split into train and test sets
train_size = int(len(data_model) * 0.67)
test_size = len(data_model) - train_size
#train, test = data_model[0:train_size,:], data_model[train_size:len(data_model),:]
train = data_nonzero[data_nonzero.segment_id.isin(train_ids)][['target']+SEG_CHARS].values
test = data_nonzero[data_nonzero.segment_id.isin(test_ids)][['target']+SEG_CHARS].values
print(len(train), len(test))
```

[PERSISTED OUTPUT]

(302720, 129664)

[CODE FOR OUTPUT]

```
# create and fit the LSTM network
mod
```

[Middle of retained excerpt omitted from the print edition.]

```
edict on all segments
data_model['risk_score'] = test.rundict['RF_base']['m_fit'].predict_proba(data_model[features])[:,1]
data_model.to_csv('seg_with_risk_score.csv', index=False)
```

E-EC0B19537E · notebooks/python2/open_street_map.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

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Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
[CODE FOR OUTPUT]
node_counts = {}
dup_nodes = {}
# Any node that's shared by more than one way is an intersection

non_named_count = 0
way_lines = []
unnamed_lines = []
service_lines = []
for way in elements:
    if way['type'] == 'way':

        coords = []
        way_nodes = way['nodes']

        prev = None
        for i in range(len(way_nodes)):
            n = way_nodes[i]
            if prev:
                coords.append((
                    prev['lon'], prev['lat']),
                    (nodes[n]['lon'], nodes[

[Middle of retained excerpt omitted from the print edition.]

st
#ways = [x for x in elements if x['type'] == 'way']

# Turn the nodes into key value pairs where key is the node id
nodes = {x['id']: x for x in elements if x['type'] == 'node'}
```


E-F313830786 · notebooks/general_analysis.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
[CODE FOR OUTPUT]
items = json.load(open(PROCESSED_DATA_FP + 'crash_joined.json'))
crash_list, crashes_by_id = util.group_json_by_location(items)
print(str(len(crash_list)) + " crashes found")

# Crashes by id is a dict where the key is the segment id. It contains the count of crashes at that
# intersection
# Can also contain other information about this location, if called with arg otherfields. In this case,
# we're
# Looking at time
sorted = [value['count'] for key,value in crashes_by_id.items() if key != '']
sort

[Middle of retained excerpt omitted from the print edition.]

ue in results.items():
    print(str(key) + ':' + str(float(value['crash'])/(float(value['no_crash'] + value['crash']))) +
" out of " + str(value['no_crash'] + value['crash']))
```

E-HIS-574EA43B97 · git:6b22e7085cfbb30e9a84e5338898141377d0a01c

Epistemic status: OBSERVED · **Kind:** git commit

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Commit: 6b22e7085cfbb30e9a84e5338898141377d0a01c
Authored: 2019-10-08T14:31:23-04:00
Subject: Remove risk_map from pipeline
Body: Not recorded
Changed files (1):
M      src/pipeline.py
```

E-HIS-5E472FA611 · git:56284d0f20d072ffd1cc2b05efd2a400f9d1655e

Epistemic status: OBSERVED · **Kind:** git commit

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Commit: 56284d0f20d072ffd1cc2b05efd2a400f9d1655e
Authored: 2019-08-28T17:11:50-04:00
Subject: added split column capability to model
Body: Not recorded
Changed files (2):
M      src/models/train_model.py
M      src/pipeline.py
```

E-HIS-63857821EE · git:tag/v2.0

Epistemic status: OBSERVED · **Kind:** git tag

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Tag: v2.0
Commit: 37de9d9855c1387a66cea4ce777052cd032d4ee7
Created: 2019-03-20T17:52:20-04:00
Subject: Update readme to new slack channel name
```

E-HIS-6E325D0345 · git:9a5203a5ec77d1d53afb361555398aa122db8a1e

Epistemic status: OBSERVED · **Kind:** git commit

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Commit: 9a5203a5ec77d1d53afb361555398aa122db8a1e
Authored: 2020-06-25T17:05:21-04:00
Subject: fix to model use appropriate featureset
Body: Not recorded
Changed files (1):
M      src/models/train_model.py
```

E-HIS-A19507381C · git:434780a182b1fa7d1908454423b932bb8143062c

Epistemic status: OBSERVED · **Kind:** git commit

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Commit: 434780a182b1fa7d1908454423b932bb8143062c
Authored: 2019-09-08T18:24:54-04:00
Subject: Update make_preds_viz + pipeline to generate postprocessed files for split_columns
Body: Not recorded
Changed files (2):
M      src/data/make_preds_viz.py
M      src/pipeline.py
```

E-HIS-C55CAB55D7 · git:tag/v1.2

Epistemic status: OBSERVED · **Kind:** git tag

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Tag: v1.2
Commit: c79a562eea13f1bc8532cd40a04113ff48c7ab0
Created: 2018-08-12T11:32:26-04:00
Subject: Merge pull request #160 from Data4Democracy/fix_bug
```

E-HIS-CD26C02968 · git:tag/v1.1

Epistemic status: OBSERVED · **Kind:** git tag

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Tag: v1.1
Commit: 26d8299c5c203a5437ca2b2db7a076d7cc3b6f69
Created: 2018-06-06T20:40:19-04:00
Subject: Merge pull request #117 from Data4Democracy/docker
```

E-HIS-D3E418A597 · git:9fb8d5d976e594b3c339048434a0aa0edbaaed45

Epistemic status: OBSERVED · **Kind:** git commit

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Commit: 9fb8d5d976e594b3c339048434a0aa0edbaaed45
Authored: 2021-12-30T10:51:10-05:00
Subject: adaptation for multi feature doc fixes and segment format fix
Body: Not recorded
Changed files (11):
M    src/data/config.py
M    src/data/create_segments.py
M    src/data/tests/test_config.py
M    src/data/util.py
M    src/data_standardization/standardization_util.py
M    src/data_standardization/standardize_crashes.py
M    src/data_standardization/standardize_point_data.py
M    src/data_standardization/tests/test_standardization_util.py
M    src/data_standardization/tests/test_standardize_point_data.py
M    src/features/make_canon_dataset.py
M    src/features/tests/test_make_canon.py
```

E-HIS-EFAA393053 · git:6edc5c8c92858d614450d10f12407e2b25bc9950

Epistemic status: OBSERVED · **Kind:** git commit

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Commit: 6edc5c8c92858d614450d10f12407e2b25bc9950
Authored: 2019-12-19T19:20:13-05:00
Subject: Added new city config for NYC. Ran pipeline and tested the web app, works but does not show
modes yet..
Body: Not recorded
Changed files (1):
A    src/config/config_nyc.yml
```

E-HIS-FCF562DDCC · git:17234a1ca127727b27df67e5c8d143144ab4f4e1

Epistemic status: OBSERVED · **Kind:** git commit

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Commit: 17234a1ca127727b27df67e5c8d143144ab4f4e1
Authored: 2019-08-14T13:00:13-04:00
Subject: Make canon dataset updates
Body: Not recorded
Changed files (3):
M      src/features/make_canon_dataset.py
M      src/features/tests/data/processed/crash_joined.json
M      src/features/tests/test_make_canon.py
```

Bóveda processing metadata

Technical audit diagnostics are retained as a subordinate appendix and do not compete with the project narrative.

TELEMETRY AVAILABILITY complete	AUDIT DURATION 63.9 s
RECONSTRUCTION RESULT accepted	ATTEMPTS / PROVIDER CALLS 1 / 1
TOTAL RECORDED TOKENS 123,245	PAID MODEL ACTIVITY DURING STORED AUDIT Yes

Attempt 1 · full reconstruction accepted	
Duration	63.8 s
Recorded tokens	123,245
Validation findings	None recorded